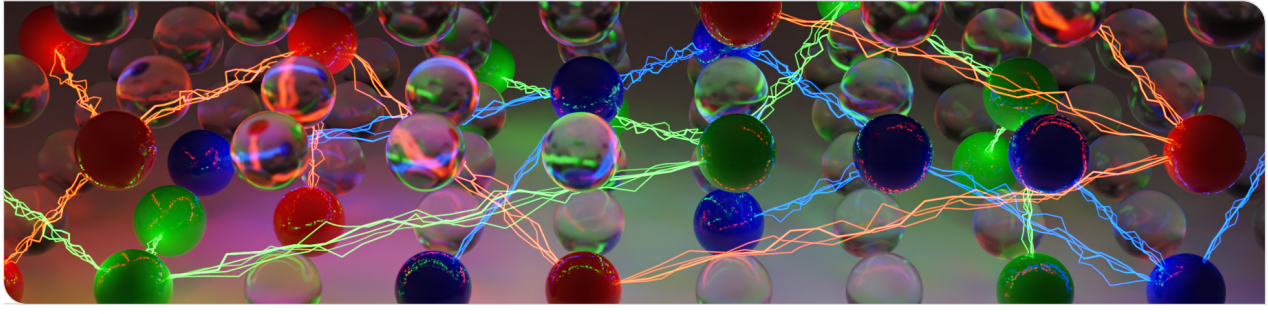
Karlsruhe Institute of Technology

Algorithm Engineering

Practical SAT Solving

Lecture 13: Proofs: Pragmatics & Parallel Production

Markus Iser, Dominik Schreiber, Tomáš Balyo | July 15, 2024



Overview

Karlsruhe Institute of Technology

Last lectures

- Background on proof systems, proof complexity
- Connections to preprocessing

This lecture

- Common state-of-the-art proof formats
- Pragmatics of proof production and checking
- Producing proofs with parallel + distributed solvers
- Beyond proof files: on-the-fly checking

Back To Basics

- Solver on unsat. formula F produces sequence of clauses $P := \langle c_1, c_2, \ldots, c_n \rangle$ with $c_n = \emptyset$
- Goal: Justify for $i = 1, \ldots, n$ that $F \models c_i$, i.e., that c_i follows from F
  - actually (in practice): that $(F \cup \bigcup_{j=1}^{i-1} c_j) \models c_i$
- **Clausal Proof** $\mathcal{P}$: Expression of P with all information needed to justify all steps

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- **Clausal Proof** $\mathcal{P}$: Expression of P with all information needed to justify all steps efficiently (?)

Approach 1: Basic Clausal Proof

- **Solving**: Solver just logs each produced c_i to a file $\Rightarrow \mathcal{P} = P$
- **Checking**: Maintain clause database B initialized as $B := F$;
  for each c_i, confirm that $B \models c_i$ and then $B := B \cup c_i$

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  for each c_i, confirm that $B \models c_i$ and then $B := B \cup c_i$
- How do we perform the confirmation step?

Efficient Redundancy Checking

The RUP Property

Given a clause set C and a clause c, we say that c has the Reverse Unit Propagation (RUP) property iff unit propagation on $(C \cup \{\overline{c}\})$, where $\overline{c} := \{\neg l \ : \ l \in c\}$, produces the empty clause.

- Is a clause c with RUP property w.r.t. a checker's clause set B a sound addition to B?
  - Yes: $B \wedge \bigwedge_{l \in c} \neg l$ is unsatisfiable $\rightarrow$ No way to satisfy B without satisfying c $\rightarrow$ $B \models c$

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- What kinds of clauses have the RUP property?
  - Conflict clauses from CDCL
  - Clauses arising from many pre– and inprocessing techniques (variable elimination, subsumption, vivification, ...)
  - Actually, all clauses produced by out-of-the-box CaDiCaL

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  - Conflict clauses from CDCL
  - Clauses arising from many pre– and inprocessing techniques (variable elimination, subsumption, vivification, ...)
  - Actually, all clauses produced by out-of-the-box CaDiCaL
- What kinds of clauses do not have the RUP property?
  - Extended Resolution steps
  - Propagation Redundancy (PR) clauses
  - ...

RUP Proof Checking [1]

Approach 2: RUP Proof

Solving: Solver just logs each produced c_i to a file $\Rightarrow \mathcal{P} = P$.

Checking:

$B := F$
for $i = 1, \ldots, n$:
    propagate $\neg l$ in B for each $l \in c_i$
    if propagation in B does *not* yield the empty clause:
        return ERROR
    undo propagations in B
    $B := B \cup c_i$
return VALIDATED

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Checking complexity:

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Checking complexity: $\mathcal{O}(|B|)$ per step $\Rightarrow \mathcal{O}(|P|^2)$ for $|P| \gg |F|$
Checking space usage:

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Checking complexity: $\mathcal{O}(|B|)$ per step $\Rightarrow \mathcal{O}(|P|^2)$ for $|P| \gg |F|$
Checking space usage: $\mathcal{O}(|F| + |P|)$

How to improve on both?

From RUP to DRUP [2]

Actually, a solver also deletes clauses. $\Rightarrow$ Put deletion information in the proof!

- $\mathcal{P} = (o_1, \ldots, o_N)$ where $o_i = (op_i, c_i)$
  - $op_i \in \{\texttt{add}, \texttt{delete}\}$
  - `delete`: c_i is a clause added by some o_j, $j < i$, and not `deleted` by any o_k, $j < k < i$
  - multi-set semantics possible where a clause may be added (+ deleted) multiple times

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Formula:

$$
\begin{aligned}
& x_1 \vee \neg x_2 \\
\wedge\ & x_2 \vee \neg x_4 \\
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\wedge\ & \neg x_1 \vee \neg x_3 \\
\wedge\ & x_1 \vee \neg x_3 \\
\wedge\ & \neg x_1 \vee x_3 \\
\wedge\ & x_1 \vee x_3 \vee \neg x_4 \\
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\end{aligned}
$$

Proof:

$$
\begin{aligned}
&\texttt{add}\ \neg x_3 \\
&\texttt{add}\ x_1 \vee x_2 \\
&\texttt{add}\ \neg x_1 \\
&\texttt{del}\ \neg x_3 \\
&\texttt{add}\ x_2 \vee x_3 \vee \neg x_4 \\
&\texttt{add}\ x_1 \vee x_2 \vee x_3 \\
&\texttt{add}\ \emptyset
\end{aligned}
$$

DRUP

Approach 3: DRUP (Deletion RUP) Proof

$B := F$
for $i = 1, \ldots, N$:
    if $op_i =$ `delete`:
        $B := B \setminus c_i$
        continue
    propagate $\neg l$ in B for each $l \in c_i$
    ... // continue as in RUP Proof

Correctness:

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Space usage:

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Complexity: $\mathcal{O}(|P| \times M)$ where M is the max. volume of present clauses during solving
Space usage: $\mathcal{O}(M)$ $\Rightarrow$ "fits into RAM if solving fits into RAM"

Can we further improve running time?

From DRUP to LRUP [3]

Idea: Enrich proof to accelerate unit propagation (UP) of $\overline{c_i}$ through B

- $\mathcal{P} = (o_1, \ldots, o_N)$ where $o_i = (\texttt{add}, id_i, c_i, d_i)$ **or** $o_i = (\texttt{delete}, id_i)$
  - $id_i \in \mathbb{N}^+$, $d_i = \langle d_{i1}, \ldots, d_{ik_i} \rangle$ where $d_{ij} \in \mathbb{N}^+$, $k_i \in \mathbb{N}^+$
- With d_i, solver references earlier clauses which UP needs to look at to arrive at the empty clause
  - for $1 \leq j < k_i$, clause # d_{ij} (i.e., the clause referred to by d_{ij}) must break down into a unit
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Formula:

(1) $x_1 \vee \neg x_2$
(2) $x_2 \vee \neg x_4$
(3) $x_1 \vee x_2 \vee x_4$
(4) $\neg x_1 \vee \neg x_3$
(5) $x_1 \vee \neg x_3$
(6) $\neg x_1 \vee x_3$
(7) $x_1 \vee x_3 \vee \neg x_4$
(8) $x_1 \vee x_3 \vee x_4$

DRUP Proof:

add $\neg x_3$
add $x_1 \vee x_2$
add $\neg x_1$
del $\neg x_3$
add $x_2 \vee x_3 \vee \neg x_4$
add $x_1 \vee x_2 \vee x_3$
add $\emptyset$

LRUP Proof:

add (9) $\neg x_3$ (5, 4)
add (10) $x_1 \vee x_2$ (3, 2)
add (11) $\neg x_1$ (6, 9)
del (9)
add (12) $x_2 \vee x_3 \vee \neg x_4$ (7, 11)
add (13) $x_1 \vee x_2 \vee x_3$ (8, 12)
add (14) $\emptyset$ (11, 10, 1)

LRUP

Approach 4: LRUP (Linear RUP) Proof Checking

$B := F$
for $i = 1, \ldots, N$:
  if $op_i =$ `delete`:
    $B := B \setminus \{\# id_i\}$ // delete clause referred to by id_i
    continue
  $U := \{\overline{l} \ : \ l \in c_i\}$
  for $j = 1, \ldots, k-1$:
    assert: clause $\# d_{ij}$ under U becomes a unit clause $\{u\}$ // returns **ERROR** upon failure
    $U := U \cup \{u\}$
  assert: clause $\# d_{ik_i}$ under U becomes the empty clause // returns **ERROR** upon failure
  $B := B \cup \{c_i\}$ // confirmed: $B \cup \{\overline{c_i}\} \models \emptyset$
return VALIDATED

$\Rightarrow$ Larger proofs but much more efficient checking (often $10\times$ or more)

From (D|L)RUP to (D|L)RAT [4]

Resolution Asymmetric Tautology (recap)

Clause c has the *Resolution Asymmetric Tautology* (RAT) property in F w.r.t. literal $x \in c$
iff every resolvent $c' \in \{c \otimes_x \tilde{c} \mid \tilde{c} \in F_{\overline{x}}\}$ has the RUP property in F.

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"Only" requiring each clause $c_i \in P$ to have the RAT property (rather than RUP) allows for stronger proofs!

- For RAT clause c, $F \cup c$ is satisfiability-preserving to F but may be not equivalent to F
- Allows to express satisfiability-preserving transformations like variable addition
- As powerful as Extended Resolution

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How to incorporate RAT into proof checking?

- DRUP $\rightarrow$ **DRAT**: For each added clause c_i, find pivot literal $x \in c_i$ and confirm that c_i is RAT in B w.r.t. x
  - Convention: 1st literal of c_i must be valid pivot
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  - Find all resolvents, check RUP for every one of them
- LRUP $\rightarrow$ **LRAT**: Additions $(\texttt{add}, id_i, c_i, d_i, r_i)$ with $r_i = \langle r_{i1}, \ldots, r_{im_i} \rangle$, $m_i \in \mathbb{N}_0$
  - Each r_{ij} references a clause $\tilde{c}$ and the required RUP steps for $c' = c_i \otimes_x \tilde{c}$ (like d_i for c_i in pure RUP)
  - Still need to internally maintain occurrences of each literal to check that all $\tilde{c}$ are covered

Proof (File) Formats: DRAT and LRAT

DIMACS CNF

```
p cnf 4 8
 1 -2    0
 2 -4    0
 1  2  4 0
-1 -3    0
 1 -3    0
-1  3    0
 1  3 -4 0
 1  3  4 0
```

DRAT proof

```
  -3       0
   1  2    0
  -1       0
d -3       0
   2  3 -4 0
   1  2  3 0
            0
```

LRAT proof

```
9     -3       0 5 4   0
10     1  2    0 3 2   0
11    -1       0 6 9   0
11 d  9        0
12     2  3 -4 0 7 11  0
13     1  2  3 0 8 12  0
14             0 11 10 1 0
```

These proofs only feature RUP additions. In an LRAT addition, each r_{ij} is written as the negated ID of $\tilde{c}$ followed by IDs for the RUP steps of c'.

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Other formats:

- FRAT: Compromise between DRAT and FRAT at the developer's discretion [5]
- DPR, LPR: Propagation Redundancy (PR) reasoning [6]
- VeriPB: Pseudo-Boolean reasoning [6]

Note: formally verified checkers are available for all these formats (sometimes translation-based)

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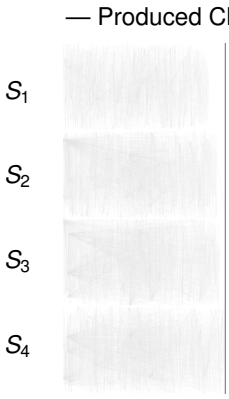
2023: LRAT-based proofs from clause-sharing solvers [9]

- Globally unique clause IDs without communication
  – for o original clauses and p solver threads, the i-th thread assigns clause IDs $o + i + kp$ ($k \in \mathbb{N}_0$)
- After solving, rewind the procedure, using "hints" of LRAT to trace dependencies of empty clause
- Funnel all clauses marked as required into a single, ordered proof file

Distributed Proof Production (1/2) [9]



Random 3-SAT formula, 180 variables. 4 notebook cores $\times$ 1.7 s. 300k dependencies (w/o orig. clauses).

Solving: Each thread derives and shares clauses, logs to partial proof file

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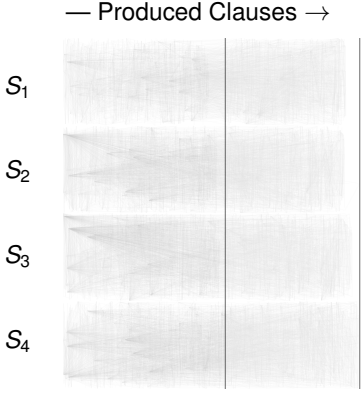

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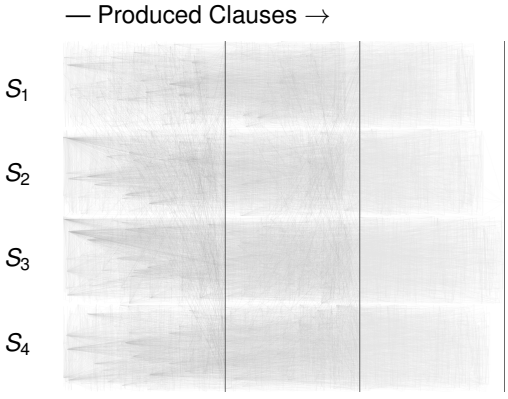

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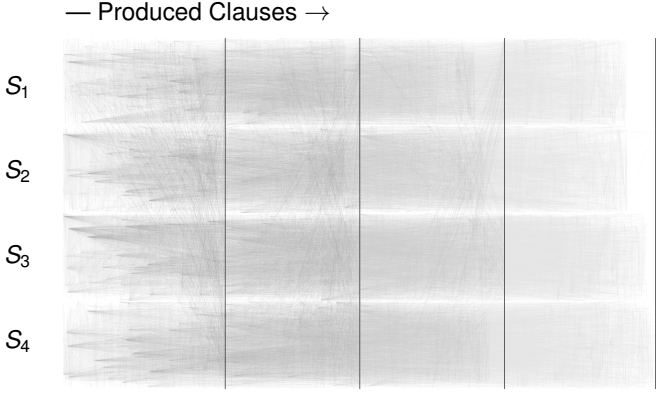

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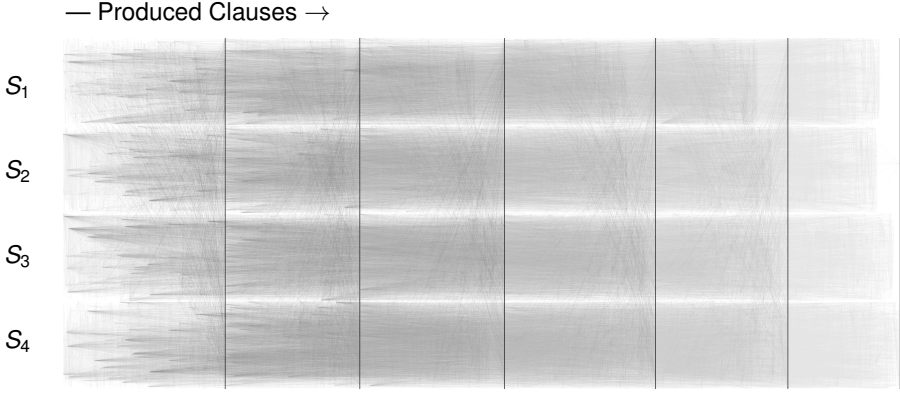

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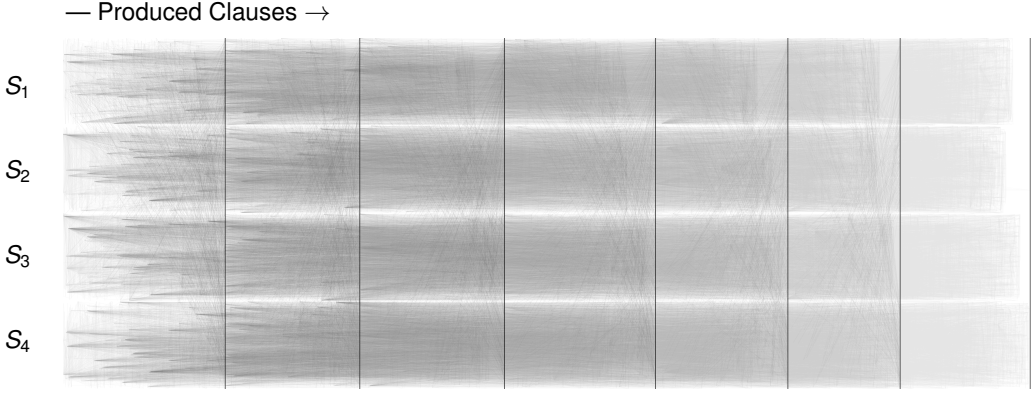

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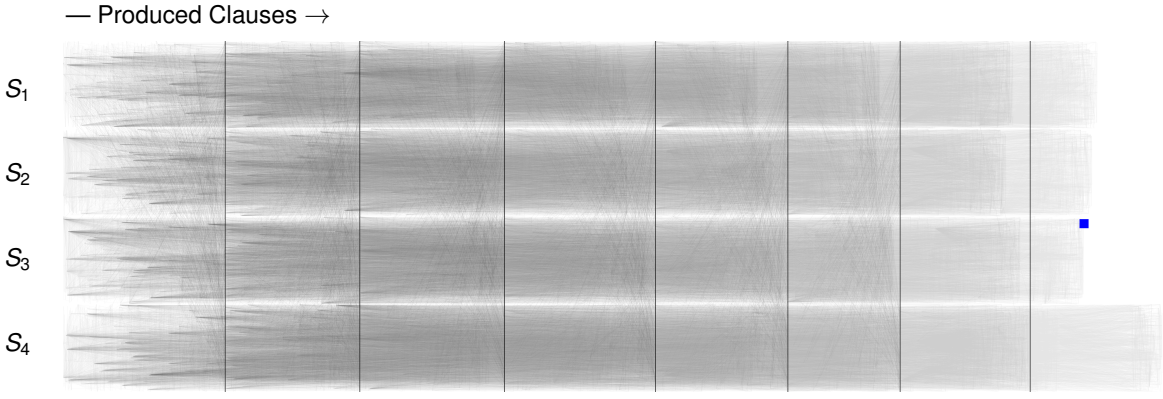

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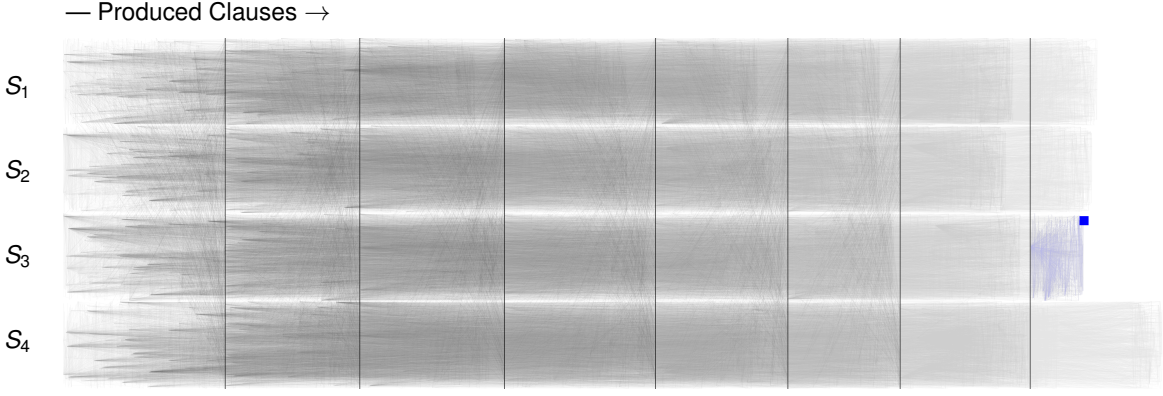

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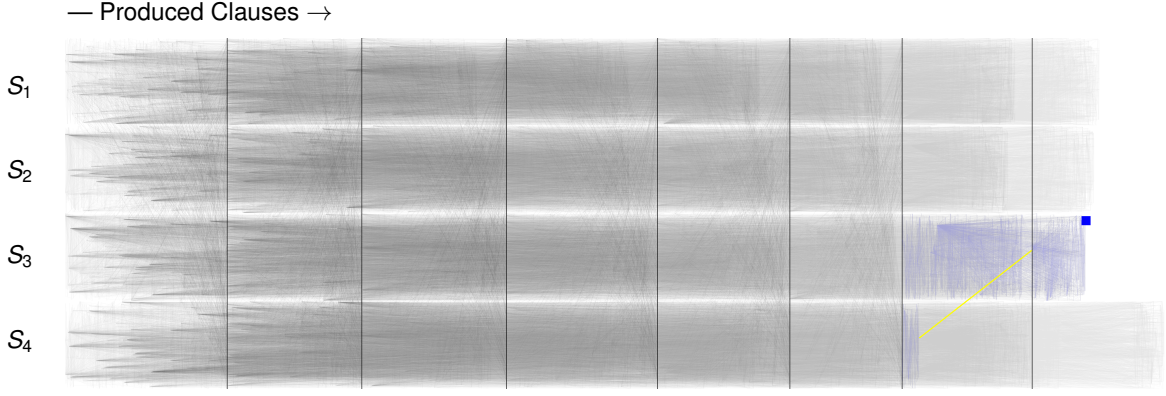

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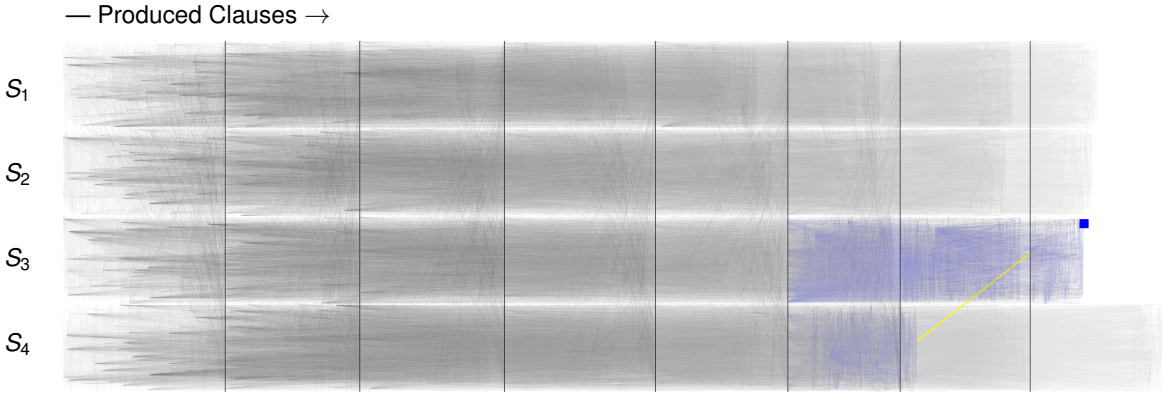

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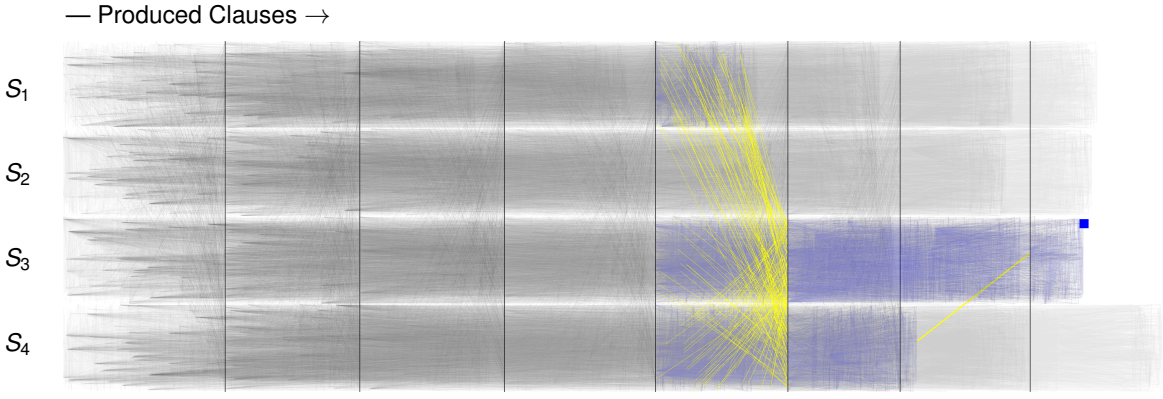

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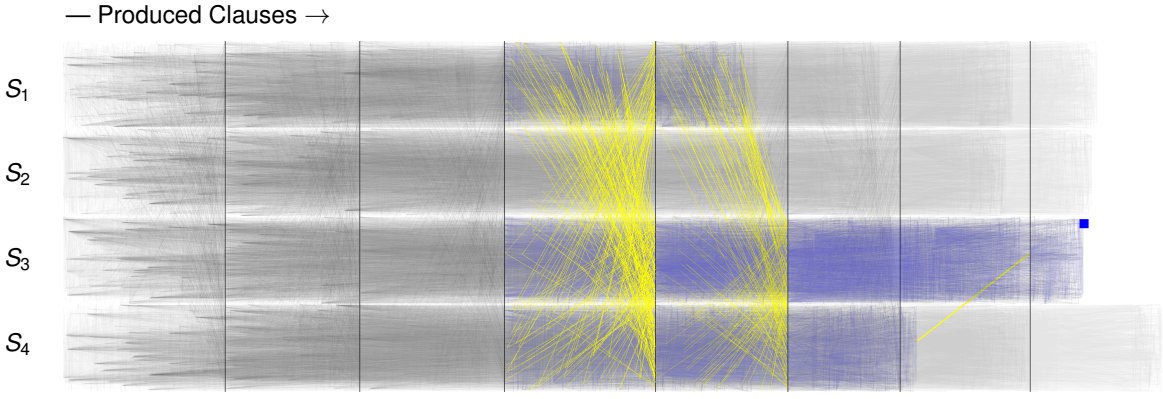

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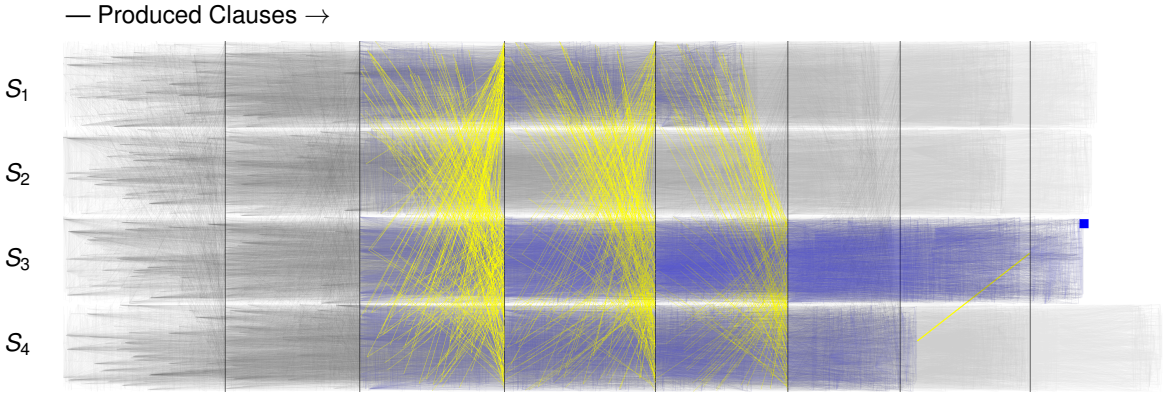

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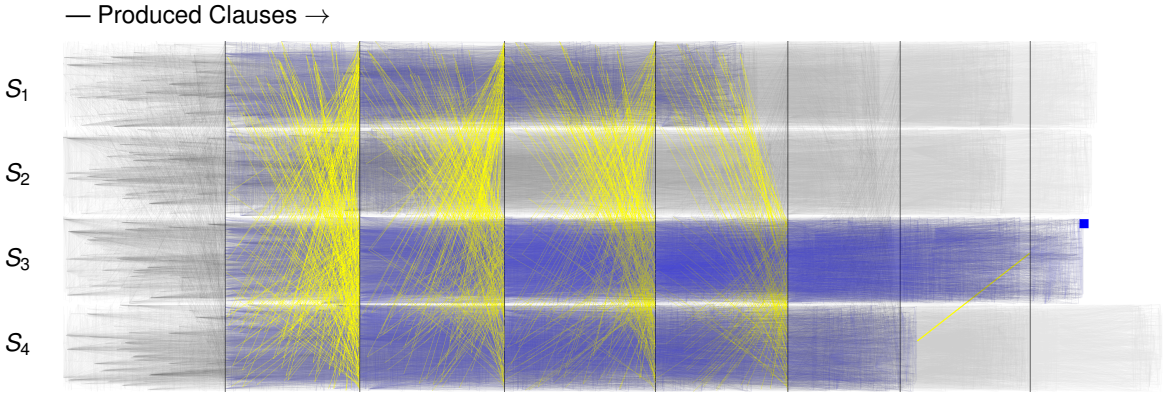

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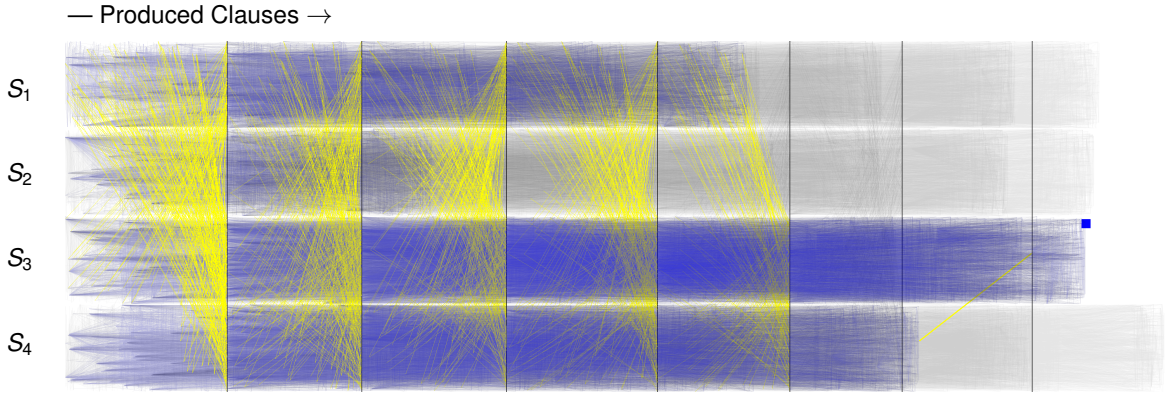

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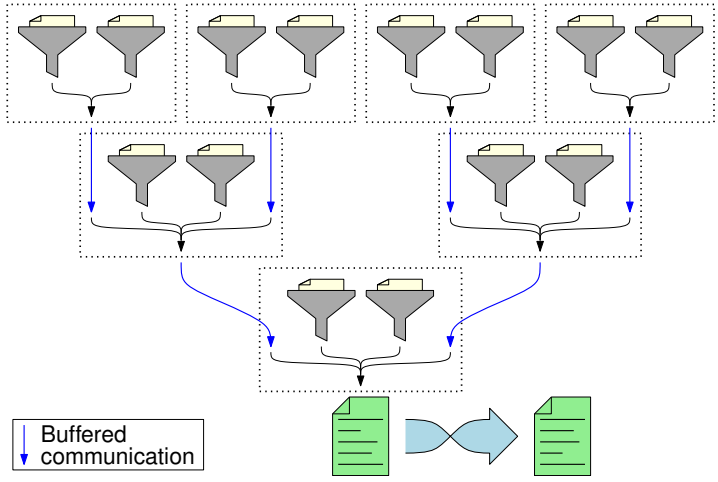
Distributed Proof Production (2/2) [9]

Merging:

- Funnel required clause additions, still in reverse order, into singular proof file
- Hierarchical merging along tree
- "Root process" writes output to file
  - Seeing an ID d_{ij} for the first time?
    $\Rightarrow$ write deletion of d_{ij} before writing the current statement!
  - Finally: Invert lines of proof file



Distributed Proof Production: Discussion

Results: (Latest setup, JAR'24 submission)

- Mean speedup of proof-emitting solver @ 1520 cores over sequential solver (solving times only): 17.5
  - Mean speedup of best MallobSat @ 1520 cores over sequential solver: 26.9
- On average, assembling and checking a proof takes $\approx 3\times$ solving time
  - Mean overhead of DRAT proof checking over sequential solving: $\approx 1\times$
- Pruning irrelevant clause additions reduces proof size by ≈ 30–$40\times$
- LRAT proof size: median 3.1 GB, mean 11.6 GB, maximum 233.9 GB

Bottleneck:

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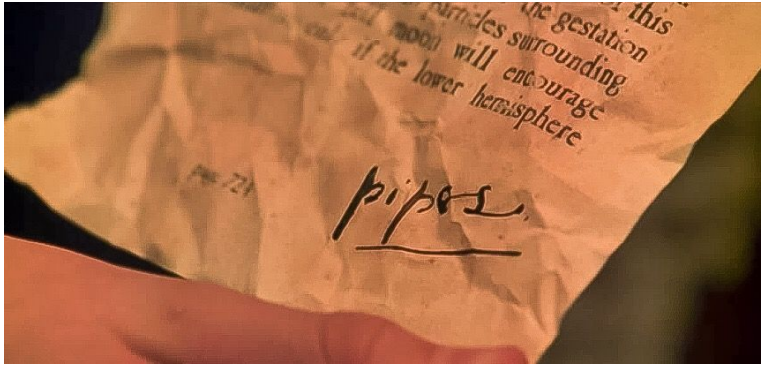
Bottleneck: Assembly and validation of a monolithic proof

- Proof creation throttled by I/O bandwidth at final process
- Checking can take very long
- The assembled proof's "corridor" of active clauses may no longer fit into RAM

Can we do better?

Hermione's Answer to More Scalable Trusted Solving

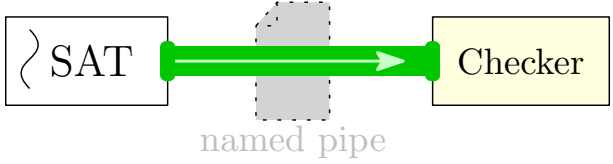


https://i.pinimg.com/originals/1b/3d/b6/1b3db639721eeafb188a3cc3060ff58b.jpg

Beyond Monolithic Proof Files

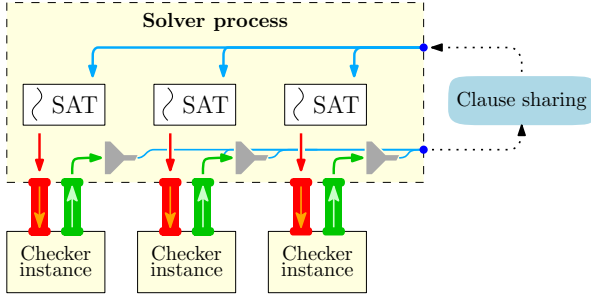
Marijn Heule: Since LRUP checking is so efficient, we can feasibly do it in realtime!

- Solver streams proof output into a pipe (UNIX special file)
- Checker reads proof from pipe and checks it on-the-fly
  – checking is done as soon as solving is done!
- Almost no slowdown when running solver and checker on two hardware threads of the same core
- No disk I/O required, same program code as with normal files (execute `mkfifo` beforehand)
- Does not yield a persistent artifact to validate by independent parties



Clause-Sharing Solving with on-the-fly Checking [10]

- One checker per solver thread
- Incoming shared clauses are forwarded to the checker without (LRUP) checking



github.com/domschrei/impcheck

Clause-Sharing Solving with on-the-fly Checking [10]

- One checker per solver thread
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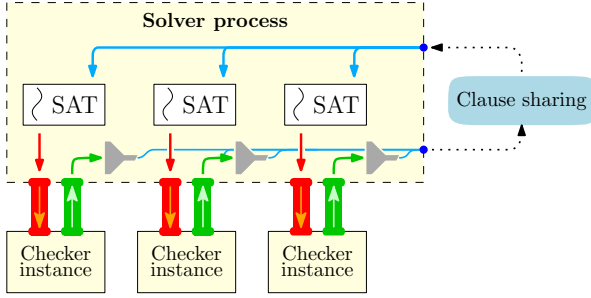
- How do we account for incorrect shared clauses?



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- One checker per solver thread
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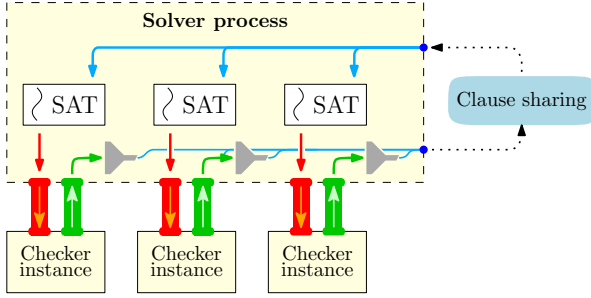
- Checkers sign each successfully checked clause c with 128-bit signature based on shared secret K:

  $\mathcal{S}_K(c) = H_K(id(c) \,||\, c \,||\, \mathcal{S}_K(F))$

  H_K: Message Authentication Code (MAC), specifically **SipHash**



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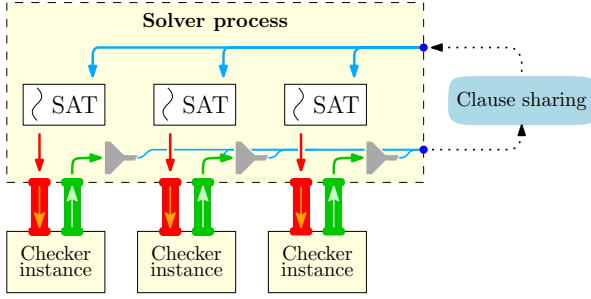
- Incoming clause c is forwarded to checker together with $\mathcal{S}_K(c)$

  $\Rightarrow$ Checker can validate that another checker has signed the clause!



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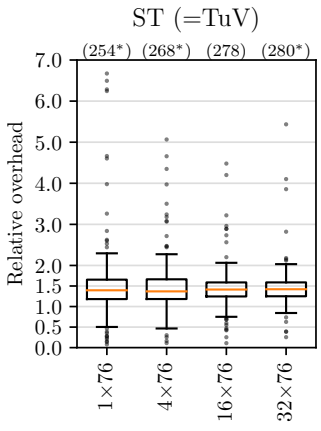
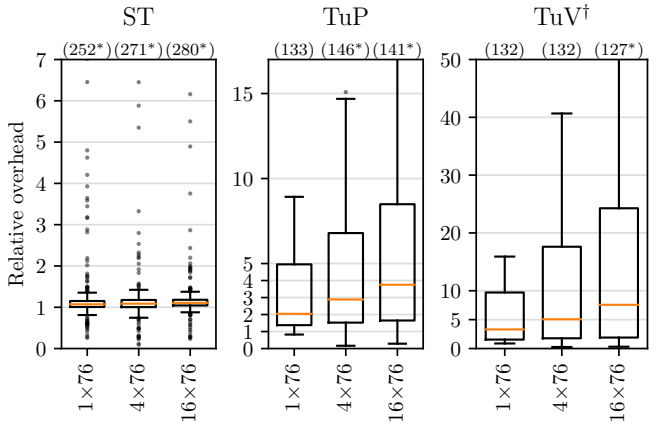
Clause-Sharing Solving + Checking: Scalability [10]

On-the-fly checking



Monolithic proofs



Overhead relative to proof-free solving time · ST: Solving time · TuP: Time until Proof present · TuV: Time until Validation done
76-core nodes · †Data extrapolated · *some data outside of displayed domain

Wrap-Up

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  - Unclear if / how well this works for actual LRAT (not LRUP) derivations
    – especially for clause-sharing solving
- **Right now:** Rise of new proof formats (PR, PB, . . .) promising shorter proofs for some problems [6]
  – DRAT / LRAT is technically just as powerful but relies on variable addition for most powerful proofs
  - huge decision space, difficult to find a short proof
  - But: recent success in effective structured variable addition [11]

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